CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 14: Linear Equations in One Variable

Topic-wise short notes with solved examples

Solving linear equations in one variable, including equations with brackets and fractions, and using equations to solve word problems on numbers, ages, digits, money and geometry.

1 Equations

- An **equation** is a statement that two expressions are equal. It has a left-hand side (L.H.S.) and a right-hand side (R.H.S.).
- A **linear equation in one variable** has one variable and its highest power is 1. Its general form is $ax + b = 0$, with $a \neq 0$.
- The **solution (root)** is the value of the variable that makes both sides equal. A linear equation has exactly one solution.
- **Rules**: the same number may be added to, subtracted from, multiplied into or divided into both sides (not dividing by zero).
- **Transposing**: a term moved to the other side changes its sign. A multiplier becomes a divisor and a divisor becomes a multiplier.
- Always **verify** by substituting the solution in the original equation.

2 Simple equations

Example 2.1 Solve $5x - 7 = 18$.

SOLUTION

$5x = 25$

$x = 5$

Example 2.2 Solve $7x + 4 = 3x - 12$.

SOLUTION

$7x - 3x = -12 - 4$, so $4x = -16$

$x = -4$

Example 2.3 Solve $\frac{x}{3} - 2 = 5$.

SOLUTION

$\frac{x}{3} = 7$

$x = 21$

3 Equations with brackets

- Remove the brackets, collect the variable terms on one side and the constants on the other, then simplify.

Example 3.1 Solve $3(x - 2) + 5 = 2(x + 4)$.

SOLUTION

$3x - 6 + 5 = 2x + 8$, so $3x - 2x = 8 + 1$

$x = 9$

Example 3.2 Solve $5(2x - 1) - 3(x + 4) = 4$.

SOLUTION

$10x - 5 - 3x - 12 = 4$, so $7x = 21$

$x = 3$

4 Equations with fractions

- Multiply every term by the **LCM of the denominators** to clear the fractions.
- Put brackets round a numerator with more than one term before multiplying.

Example 4.1 Solve $\frac{x}{2} + \frac{x}{3} = 10$.

SOLUTION

Multiply by 6: $3x + 2x = 60$

$x = 12$

Example 4.2 Solve $\frac{2x - 1}{3} - \frac{x + 2}{4} = 1$.

SOLUTION

Multiply by 12: $4(2x - 1) - 3(x + 2) = 12$, so $8x - 4 - 3x - 6 = 12$ and $5x = 22$.

$x = \frac{22}{5}$

Example 4.3 Solve $\frac{3x+5}{2x-1} = \frac{4}{3}$.

SOLUTION

Cross-multiply: $3(3x+5) = 4(2x-1)$, so $9x + 15 = 8x - 4$.

$x = -19$

Example 4.4 Solve $0.5x + 0.2 = 0.3x + 1.4$.

SOLUTION

Multiply by 10: $5x + 2 = 3x + 14$

$x = 6$

5 Word problems: method

- Read the problem and decide what is unknown. Call it x.
- Write the other quantities in terms of x.
- Form an equation from the condition given.
- Solve it, answer in words, and check against the problem.

Words	Algebra
5 more than a number	$x + 5$
3 less than twice a number	$2x - 3$
Three consecutive numbers	$x, x+1, x+2$
Consecutive even (or odd) numbers	$x, x+2, x+4$
Two numbers in the ratio $3:5$	$3x, 5x$
Two-digit number, digits x (tens) and y (units)	$10x + y$

6 Numbers

Example 6.1 The sum of three consecutive numbers is 87. Find them.

SOLUTION

$x + x + 1 + x + 2 = 87$, so $3x = 84$

28, 29, 30

Example 6.2 One number is 3 times another, and their difference is 36. Find them.

SOLUTION

$3x - x = 36$

18 and 54

Example 6.3 A number is as much greater than 21 as it is less than 71. Find it.

SOLUTION

$x - 21 = 71 - x$

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Example 6.4 The sum of the digits of a two-digit number is 11. The number formed by reversing the digits is 27 more than the original. Find the number.

SOLUTION

Let the units digit be x; the tens digit is $11 - x$. Number $= 10(11 - x) + x = 110 - 9x$; reversed $= 10x + 11 - x = 9x + 11$.

$9x + 11 - (110 - 9x) = 27$, so $18x = 126$ and $x = 7$.

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7 Ages and money

Example 7.1 A father is 30 years older than his son. In 5 years he will be 3 times as old as his son. Find their present ages.

SOLUTION

Son x, father $x + 30$. $x + 35 = 3(x + 5)$, so $2x = 20$.

Son 10 years, father 40 years

Example 7.2 The ages of A and B are in the ratio $5 : 7$. Four years ago the ratio was $2 : 3$. Find their present ages.

SOLUTION

$\frac{5x - 4}{7x - 4} = \frac{2}{3}$, so $15x - 12 = 14x - 8$ and $x = 4$.

20 years and 28 years

Example 7.3 A purse has ₹5 and ₹2 coins, 40 coins in all, worth ₹140. How many coins of each kind are there?

SOLUTION

₹5 coins x, ₹2 coins $40 - x$: $5x + 2(40 - x) = 140$, so $3x = 60$.

20 coins of each kind

8 Geometry and speed

Example 8.1 The length of a rectangle is 5 cm more than its breadth and its perimeter is 50 cm. Find its sides.

SOLUTION

$2(x + x + 5) = 50$, so $2x + 5 = 25$

Breadth 10 cm, length 15 cm

Example 8.2 The angles of a triangle are x, $x + 20^\circ$ and $2x$. Find them.

SOLUTION

$4x + 20 = 180$

40°, 60°, 80°

Example 8.3 A man walks from home to a station at 4 km/h and returns at 6 km/h. The whole trip takes 5 hours. Find the distance.

SOLUTION

$\frac{d}{4} + \frac{d}{6} = 5$; multiply by 12: $5d = 60$

12 km

9 Common mistakes

- Forgetting to change the sign when transposing a term.
- Multiplying only some of the terms by the LCM.
- Dropping the bracket after a minus sign: $-(x + 2)$ is $-x - 2$.
- Answering with x when the question asks for another quantity (for example the father's age).
- Writing a two-digit number as xy.
- Not checking the answer in the original problem.